

▼ **Figure 46.12 Androgen-dependent male anatomy and behavior in a moose.** The male and female in this mating pair of moose (*Alces alces*) differ in both anatomy and physiology. High levels of testosterone in the male are responsible for the appearance of secondary sex characteristics, such as antlers, and for male courtship and territorial behavior.



promote specific sexual behaviors and sex drive, as well as an increase in general aggressiveness. Estrogens similarly have multiple effects in females. At puberty, estradiol stimulates breast and pubic hair development. Estradiol also influences female sexual behavior, induces fat deposition in the breasts and hips, increases water retention, and alters calcium metabolism.

Biological Sex, Gender Identity, and Sexual Orientation in Human Sexuality

A newborn baby is assigned a “biological sex” that typically reflects the genitals present at birth and the child’s chromosomes. In mammals, the Y chromosome carries a gene called *SRY* that directs development of the gonad into a testis. In XX embryos, which lack the Y and hence lack *SRY*, the gonad becomes an ovary. Although most individuals are born male or female, roughly one in 100 is intersex, having both male and female biological characteristics. For example, intersex individuals may have a nonstandard chromosome set (such as XXY) or may differ in the hormone-directed pathways that control sexual development.

Though often confused with assigned sex, gender identity is distinct and refers to a person’s internal sense of being male, female, some combination, or neither. The term *cisgender* describes a person having a gender identity in line with their assigned sex. In contrast, a *transgender* person experiences a mismatch between their gender identity and their assigned sex. Thus, for example, an individual may have an assigned sex of female, but a male gender identity.

Whereas gender identity is about a person’s self, sexual orientation identifies the gender of people to whom an individual is attracted romantically, emotionally, and sexually. Members of a human population may have a sexual

orientation that is heterosexual (straight), homosexual (lesbian or gay), bisexual, or asexual. As these descriptions make clear, human sexuality varies considerably.

Hormonal Control of the Male Reproductive System

When mammals reach sexual maturity, the sex hormones and gonadotropins have essential roles in gametogenesis. In exploring this hormonal control of reproduction, we’ll begin with the relatively simple system found in males.

In directing spermatogenesis, FSH and LH act on two types of cells in the testis (**Figure 46.13**). FSH stimulates *Sertoli cells*, located within the seminiferous tubules, to nourish developing sperm (see Figure 46.11). LH causes *Leydig cells*, scattered in connective tissue between the tubules, to produce testosterone and other androgens, which promote spermatogenesis in the tubules.

Two negative-feedback mechanisms control sex hormone production in males (see Figure 46.13). Testosterone regulates the blood concentration of GnRH, FSH, and LH through inhibitory effects on the hypothalamus and anterior pituitary. In addition, *inhibin*, a hormone that in males is produced by Sertoli cells, acts on the anterior pituitary gland to reduce FSH secretion. Together, these negative-feedback circuits maintain the level of androgen in the normal range.

Leydig cells have other roles besides producing testosterone. They in fact secrete small quantities of many other hormones and local regulators, including oxytocin, renin, angiotensin, corticotropin-releasing factor, growth factors, and prostaglandins. These signals coordinate the activity of reproduction with growth, metabolism, homeostasis, and behavior.

▼ **Figure 46.13 Hormonal control of the testes.**

